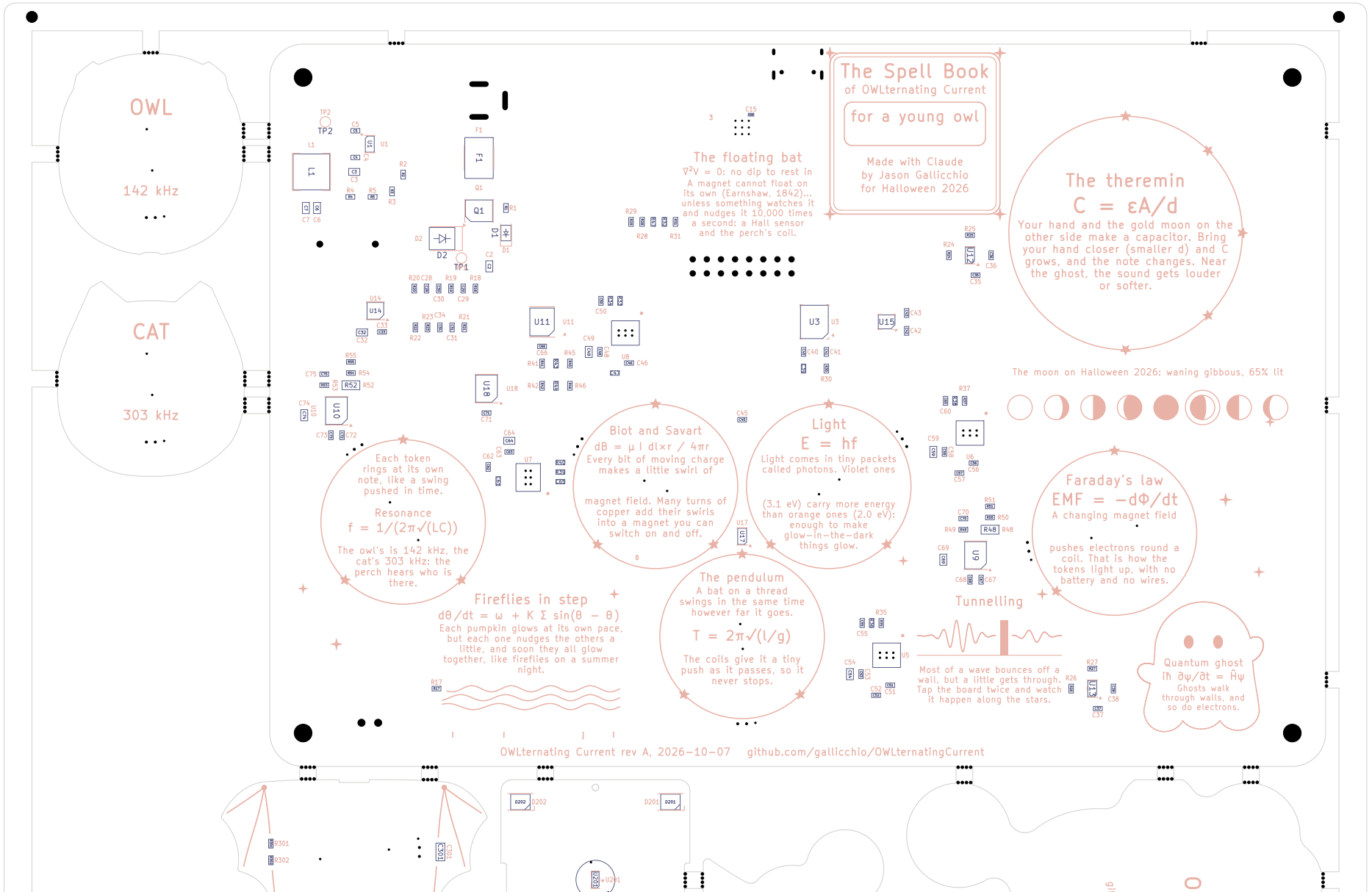

OWL
142 kHz
CAT
303 kHz
The Spell Book
of OWLternating Current
for a young owl
Made with Claude
by Jason Gallicchio
for Halloween 2026
The floating bat
$\nabla^2 V = 0$: no dip to rest in
A magnet cannot float on its own (Earnshaw, 1842)... unless something watches it and nudges it 10,000 times a second: a Hall sensor and the perch's coil.
The theremin
$C = \varepsilon A/d$
Your hand and the gold moon on the other side make a capacitor. Bring your hand closer (smaller d) and C grows, and the note changes. Near the ghost, the sound gets louder or softer.
The moon on Halloween 2026: waning gibbous, 65% lit
Each token rings at its own note, like a swing pushed in time.
Resonance
$f = 1/(2\pi\sqrt{LC})$
The owl's is 142 kHz, the cat's 303 kHz: the perch hears who is there.
Biot and Savart
$dB = \mu I\, dl \times r / 4\pi r$
Every bit of moving charge makes a little swirl of
magnet field. Many turns of copper add their swirls into a magnet you can switch on and off.
Light
$E = hf$
Light comes in tiny packets called photons. Violet ones
(3.1 eV) carry more energy than orange ones (2.0 eV): enough to make glow-in-the-dark things glow.
Faraday's law
$EMF = -d\Phi/dt$
A changing magnet field
pushes electrons round a coil. That is how the tokens light up, with no battery and no wires.
Fireflies in step
$d\theta/dt = \omega + K \Sigma \sin(\theta - \theta)$
Each pumpkin glows at its own pace, but each one nudges the others a little, and soon they all glow together, like fireflies on a summer night.
The pendulum
A bat on a thread swings in the same time however far it goes.
$T = 2\pi\sqrt{(l/g)}$
The coils give it a tiny push as it passes, so it never stops.
Tunnelling
Most of a wave bounces off a wall, but a little gets through. Tap the board twice and watch it happen along the stars.
Quantum ghost
$i\hbar\, \partial\psi/\partial t = \hat{H}\psi$
Ghosts walk through walls, and so do electrons.
OWLternating Current rev A, 2026-10-07 github.com/gallicchio/OWLternatingCurrent